ENRICHED LENSES (EXTENDED ABSTRACT)

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A bidirectional transformation is a specification of when the joint state of two systems should be regarded as consistent, together with a protocol for updating each system to restore consistency in response to a change in the other. Since Pierce et al. introduced the term *lens* for their model of bidirectional transformations for tree-structured data [5], the term has been adopted more widely to describe a menagerie of mathematical structures in applied category theory. In our talk, we will introduce the new notion of *enriched lenses* that subsumes the existing notions of *delta lenses* [4] and *weighted lenses* [8], and give examples of several bases of enrichment for which enriched cofunctors and enriched lenses have an applied interpretation. We will also provide conditions for an enriched cofunctor to be represented by a certain span of enriched functors.

Enriched lenses are a kind of morphism between *enriched categories*. Generalising from the characterisation of a delta lens as a functor and a cofunctor that are compatible with each other [1], an enriched lens is defined as a compatible enriched functor and *enriched cofunctor*.

Definition 1. Let $\mathcal{A}$ and $\mathcal{B}$ be categories enriched in a distributive monoidal category $(\mathcal{V}, \otimes, I)$. An *enriched cofunctor* $(F, \varphi)\colon \mathcal{A} \nrightarrow \mathcal{B}$ consists of a function $F\colon \operatorname{obj}(\mathcal{A}) \to \operatorname{obj}(\mathcal{B})$ and a family

$$\varphi_{a,b}\colon \mathcal{B}(Fa, b) \longrightarrow \sum_{x \in X} \mathcal{A}(a, x) \qquad\qquad X = F^{-1}\{b\}$$

of morphisms in $\mathcal{V}$, such that the following diagrams, which encode preservation of identities and composites, commute.

$$\begin{tikzcd}
I \arrow[r, "\eta_{Fa}"] \arrow[d, "\eta_a"'] & \mathcal{B}(Fa, Fa) \arrow[d, "\varphi_{a,Fa}"] \\
\mathcal{A}(a, a) \arrow[r, "\iota_a"'] & \sum_{x \in X} \mathcal{A}(a, x)
\end{tikzcd} \qquad X = F^{-1}\{Fa\}$$

$$\begin{tikzcd}
\mathcal{B}(Fa, b) \otimes \mathcal{B}(b, b') \arrow[r, "\mu_{Fa,b,b'}"] \arrow[d, "\varphi_{a,b} \otimes \mathrm{id}"'] & \mathcal{B}(Fa, b') \arrow[dddd, "\varphi_{a,b'}"] \\
\Big(\sum_{x \in X} \mathcal{A}(a, x)\Big) \otimes \mathcal{B}(b, b') \arrow[d, "\cong"'] & \\
\sum_{x \in X} \mathcal{A}(a, x) \otimes \mathcal{B}(Fx, b') \arrow[d, "\sum \mathrm{id} \otimes \varphi_{x,b'}"'] & \\
\sum_{x \in X} \mathcal{A}(a, x) \otimes \Big(\sum_{y \in Y} \mathcal{A}(x, y)\Big) \arrow[d, "\cong"'] & \\
\sum_{y \in Y} \sum_{x \in X} \mathcal{A}(a, x) \otimes \mathcal{A}(x, y) \arrow[r, "\sum\sum \mu_{a,x,y}"'] & \sum_{y \in Y} \mathcal{A}(a, y)
\end{tikzcd} \qquad \begin{array}{l} X = F^{-1}\{b\} \\ \\ Y = F^{-1}\{b'\} \end{array}$$

There are two surprises in the above definition: the distributivity requirement on the base of enrichment $\mathcal{V}$ and the indexing of the sum in the codomain of $\varphi_{a,b}$ by the fibre $F^{-1}\{b\}$ rather than all of $\operatorname{obj}(\mathcal{A})$. Both of these are essential to formulate the axiom expressing the compatibility of the cofunctor (F, φ) with the composition μ in $\mathcal{A}$ and $\mathcal{B}$.

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Proposition 2. *Let $(\mathcal{V}, \otimes, I)$ be an extensive category with a distributive monoidal structure. Then an enriched cofunctor $(F, \varphi)\colon \mathcal{A} \nrightarrow \mathcal{B}$ may be represented as a span of enriched functors.*

Proof (sketch). Given an enriched cofunctor $(F, \varphi)\colon \mathcal{A} \nrightarrow \mathcal{B}$, define an enriched category $\mathcal{C}$ with the same objects as $\mathcal{A}$, and whose hom-objects are constructed by pullback along coproduct injections:

$$\begin{tikzcd} \mathcal{C}(a, a') \arrow[r, "Q_{a,a'}"] \arrow[d, "P_{a,a'}"'] \arrow[dr, phantom, "\lrcorner", very near start] & \mathcal{B}(Fa, Fa') \arrow[d, "\varphi_{a,Fa'}"] \\ \mathcal{A}(a, a') \arrow[r, "\iota_{a'}"'] & \sum_{x \in X} \mathcal{A}(a, x) \end{tikzcd} \qquad X = F^{-1}\{Fa'\}$$

The pullback projections are components of enriched functors $P\colon \mathcal{C} \to \mathcal{A}$ and $Q\colon \mathcal{C} \to \mathcal{B}$, yielding the span representation of an enriched cofunctor (F, φ). □

Remark. Note that the enriched functor P is bijective on objects, while the enriched functor Q has the property that the morphism $[Q_{a,y}]\colon \sum_{y \in Y} \mathcal{A}(a, y) \to \mathcal{B}(Qa, b)$, where $Y = Q^{-1}\{b\}$, is invertible for each $a \in \mathrm{obj}(\mathcal{A})$ and $b \in \mathrm{obj}(\mathcal{B})$. These properties characterise amongst all spans of enriched functors those that correspond to cofunctors.

Composition of enriched cofunctors is unital and associative, thus for each distributive monoidal category $(\mathcal{V}, \otimes, I)$ we obtain a category $\mathcal{V}$-$\mathcal{C}$of of enriched categories and enriched cofunctors. Furthermore, there is a flat double category $\mathbb{C}\mathrm{of}(\mathcal{V})$ whose objects are enriched categories, whose horizontal morphisms are enriched functors, and whose vertical morphisms are enriched cofunctors. From Proposition 2, the double category $\mathbb{C}\mathrm{of}(\mathcal{V})$ is *span representable* [7].

Definition 3. An *enriched lens* $(F, \varphi)\colon \mathcal{A} \rightleftharpoons \mathcal{B}$ consists of an enriched functor $F\colon \mathcal{A} \to \mathcal{B}$ and an enriched cofunctor $(F, \varphi)\colon \mathcal{A} \nrightarrow \mathcal{B}$ such that the following diagram commutes:

$$\begin{tikzcd} & \sum_{x \in X} \mathcal{A}(a, x) \arrow[dr, "{[F_{a,x}]}"] & \\ \mathcal{B}(Fa, b) \arrow[ur, "\varphi_{a,b}"] \arrow[rr, "\mathrm{id}"'] & & \mathcal{B}(Fa, b) \end{tikzcd} \qquad X = F^{-1}\{b\}$$

Example 4. Let $\mathcal{V} = (\mathcal{S}\mathrm{et}, \times, 1)$. Then enriched categories and enriched cofunctors are ordinary categories and cofunctors, respectively. An enriched lens is precisely a *delta lens* [4].

Example 5. The category w$\mathcal{S}$et of *weighted sets* [6] has objects given by sets A equipped with a weight function $|-|\colon A \to [0, \infty]$, and morphisms given by functions $f\colon A \to B$ such that $|fa| \leqslant |a|$ for all $a \in A$. It is a distributive monoidal category with $A \otimes B$ given by the set $A \times B$ with $|(a, b)| = |a| + |b|$, monoidal unit I given by the set $\{*\}$ with $|*| = 0$, and coproducts given by disjoint union. Categories and functors enriched in w$\mathcal{S}$et are weighted categories and weighted functors. An enriched lens is precisely a *weighted lens*, recently introduced by Perrone [8].

Other bases of enrichment whose enriched categories, cofunctors, and lenses have interesting applied interpretations include:

- $(\mathcal{P}M, \subseteq, \cdot, \{e\})$, where $(M, \cdot, e)$ is a monoid and $\mathcal{P}$ is the powerset monad, whose enriched categories are closely related to generalised automata [2];
- $([0, \infty], \geqslant, +, 0)$ whose enriched categories are Lawvere metric spaces and whose enriched cofunctors are submetries[1];
- $(\{-\infty\} \cup [0, \infty], \leqslant, +, 0)$ whose enriched categories are related to project scheduling [9].

Developing an enriched analogue of split opfibrations to describe *least-change* enriched lenses, and generalising the Grothendieck construction for enriched lenses are both topics of active research.

[1]This observation is due to Callum Reader.

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